



AutoClock — Product Requirements Document (PRD)

Project	AutoClock — One-Click Daily Worklog & EOD Reporter
For	Iksula HackFest 2026 → live pilot with ~60 employees
Owner	Yogesh Mohite (Sr. QA)
Document	PRD v1.0
Goal type	Internal production tool (real users after the hackathon)
Hard rules	(1) Zero cost — only free / already-owned tools. (2) Built for the real ~60 daily users .
Status	Draft for build · HackFest: Fri 22 May 20:00 → Sat 23 May ~14:00
Companion docs	AutoClock_ERD.md (engineering), AutoClock_WorkPlan.md (execution), AutoClock_DevDoc.md (developer reference), brainstorm.md (background)

1. Executive Summary

Around 60 Iksula employees enter the **same time data three times every day** — a personal Google Sheet timesheet, Jira Clockwork worklogs ticket-by-ticket, and an end-of-day status email. AutoClock collapses that into **one entry**: the employee logs a short slot (picking project + Jira task), and at end of day one confirmed click posts the Jira worklogs, appends the Google Sheet, and drafts the EOD email. Five role-based views serve employees, leads, management, operations and an admin. The entire system runs on **free and already-owned tools** — no paid SaaS, no paid API, no paid hosting.

2. Problem Statement

Each employee, every working day:

1. Keeps a personal Google Sheet "Time sheet" (Time → Duration → Task Description).
2. Re-enters each item into **Jira Clockwork** — open ticket → gear → + → fill the Log Work dialog → save. 5–8 times.
3. Re-formats the same data into an **HTML EOD email** to their lead.

Cost of the problem: 10–15 minutes lost per person per day; forgotten hours reconstructed from memory at week-end; inconsistent Clockwork compliance; inaccurate project-costing data; QA leads and the Operations team spending time chasing missing logs.

3. Goals & North Star

North Star Metric (NSM): Average time an employee spends logging a full day — target **under 2 minutes** (from ~12 today).

ID	Supporting goal	Target
GM-01	Reduction in daily worklog time	≥ 80%
GM-02	Clockwork compliance across pilot teams	≥ 95%
GM-03	EOD email draft created for the user	100% of pilot users
GM-04	Employees actively using AutoClock after 2 weeks	≥ 80% of the ~60
GM-05	Running cost of the system	₹0 — always

4. Non-Goals (Out of Scope)

- Replacing Jira, Clockwork, Google Sheets or Gmail — AutoClock *feeds* them, it does not replace them.
- Automatic background/activity tracking, screenshots, or keystroke monitoring — explicitly excluded (trust).
- A mobile native app (a responsive web app is enough; a PWA is a future item).
- Company-wide rollout to ~500 — possible later, **not** the build target.
- Any paid tier, paid API, or paid hosting — forbidden by the hard rule.
- A standalone chatbot — the plain-text input already covers it.

5. Target Users & Roles

Role	Who	Primary need
Employee	The ~60 daily loggers (QA, dev, etc.)	Log the day in seconds; never open 3 systems
PM / QA Lead	Team leads (e.g., Priya K.)	See <i>their team's</i> hours & ticket effort; spot who hasn't logged
Management	Department heads	Org-wide utilization & trends — no ticket detail
Operations	Hours-compliance owner (e.g., Omkar P.)	Chase the weekly 40-hour fill; Fri/Mon reminders
Admin	System owner (IT)	Manage users, roles, project↔Jira mapping, integrations

6. Jobs To Be Done & User Stories

Acceptance criteria use **Given / When / Then**.

US-01 — Log a work slot. *As an Employee, I want to log a slot in ~20 seconds so I never reconstruct my day from memory.*

- Given I am logged in, When the reminder fires or I open AutoClock, Then I can pick a Project, pick the dependent Jira Task, type a description, and confirm a duration.

- Given I save a slot, Then it appears in my "Today" list immediately.

US-02 — One-click EOD sync. *As an Employee, I want one click to push my whole day to all three systems.*

- Given I have logged slots, When I click "Close My Day", Then I see a **preview** grouped by ticket.
- Given I confirm the preview, Then AutoClock creates the Jira worklogs, appends the Google Sheet row(s), and creates the Gmail EOD draft — and shows a success state per system.
- Given any system fails, Then that item is marked failed and can be retried; the others still succeed.

US-03 — Correct attribution. *As an Employee, I want my worklogs to show as **mine** in Clockwork.*

- Given I connected my own Jira account, When AutoClock posts a worklog, Then Clockwork shows it under my name (see ERD §7.1 — per-user auth).

US-04 — Edit before sync. *As an Employee, I want to fix a wrong entry before it is written.*

- Given an entry in "Today", When I edit or delete it, Then the change is reflected before "Close My Day".

US-05 — Reminder, not nag. *As an Employee, I want a gentle reminder, not constant pings.*

- Given the default cadence, Then I get a few nudges a day, not one every hour.
- Given I open Settings, Then I can change the cadence or switch to end-of-day-only.

US-06 — Team view (PM/Lead). *As a Lead, I want to see only my team's hours and ticket effort.*

- Given I am a Lead, When I open the dashboard, Then I see my team only — never other teams.
- Then I see effort by Jira ticket, a "not logged today" list, and a per-member table.

US-07 — Org view (Management). *As Management, I want utilization and trends across all teams.*

- Given I am Management, Then I see org-wide utilization, project portfolio and multi-week trends — not per-ticket detail.

US-08 — Weekly compliance chase (Operations). *As Operations, I want to chase everyone under the weekly target.*

- Given Friday 1 PM, When the check runs (cron or "Run now"), Then everyone below the leave-adjusted target gets a reminder email and the run is logged.
- Given Monday 1 PM, When the re-check runs, Then anyone who has since complied is dropped and only the rest are re-emailed.

US-09 — Leave-aware targets. *As Operations, I want the 40-hour target to account for leave.*

- Given an employee has approved leave, Then their weekly target is reduced by 8h per leave day and they are not chased for that time.

US-10 — System setup (Admin). *As Admin, I want to manage users, roles and the project↔Jira mapping.*

- Given I am Admin, Then I can add/disable users, set roles, and define each Project with its Jira key (which powers the Employee task dropdown).

US-11 — Log fully from the browser. *As an Employee, I want to do everything from the Chrome extension.*

- Given the extension popup, Then I can add an entry (project → task → describe → save) and run "Close My Day" without opening the web app.
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7. Core User Flows

Flow A — Log a slot (web or extension)

1. Reminder fires → user opens AutoClock.
2. Select **Project** → dependent **Jira Task** dropdown loads.
3. Type description → adjust duration/slot → **Save**.
4. Entry appears in "Today".

Flow B — Close My Day (the magic moment)

1. User clicks **Close My Day**.
2. AutoClock groups entries by ticket, sums durations, tidies descriptions (free parser).
3. **Preview screen** shows exactly what will be written.
4. User clicks **Confirm**.
5. Backend, using the user's own tokens: posts Jira worklogs → appends Google Sheet → creates Gmail draft.
6. Per-system success/fail shown; failed items retryable.

Flow C — Operations weekly chase

1. Fri 1 PM: cron (or "Run Friday Check Now") computes each person's weekly hours vs leave-adjusted target.
2. Under-target users get a reminder email; the run + recipients are logged.
3. Mon 1 PM: re-check; compliant users dropped; the rest re-emailed; escalate to lead if still short.

Flow D — First-time connect (pilot onboarding)

1. New user signs in to AutoClock.
2. User connects their **Jira** account (OAuth consent) and their **Google** account (OAuth consent).
3. AutoClock stores per-user tokens (encrypted) and is ready to log on their behalf.

8. Functional Requirements

ID	Requirement	Priority
FR-01	Employee can create, edit, delete a work entry (project, Jira task, description, duration, slot)	P0
FR-02	Project dropdown → dependent Jira-task dropdown (tasks scoped to the project)	P0
FR-03	Free deterministic parser: normalize durations, tidy descriptions, group by ticket	P0
FR-04	"Close My Day" preview + confirm before any external write	P0
FR-05	Post worklogs to Jira via the user's own auth → Clockwork displays them	P0
FR-06	Append a row to the user's Google Sheet timesheet	P0
FR-07	Create a Gmail EOD draft in the user's account	P0
FR-08	Per-system success/failure status + retry of failed items	P0
FR-09	RBAC — 5 roles, each restricted to its own scope	P0
FR-10	PM/Lead dashboard — team-scoped delivery metrics	P1
FR-11	Management dashboard — org-wide strategic metrics	P1

FR-12	Operations console — weekly compliance tracker + reminder queue + history	P1
FR-13	Admin console — users, roles, project↔Jira mapping, integrations, settings	P1
FR-14	Chrome extension — full entry + reminders + "Close My Day"	P1
FR-15	Configurable reminder cadence (incl. EOD-only)	P1
FR-16	Friday/Monday compliance cron + manual "Run now"	P1
FR-17	Leave/holiday calendar adjusts the weekly target	P1
FR-18	Timezone handling — IST entry ↔ UTC for Jira	P0
FR-19	All secrets/tokens encrypted at rest; never in the repo	P0
FR-20	"Reconstruct Yesterday" from Jira activity	P3 (future)
FR-21	Read all employees' worklogs from Jira (via one Browse-Projects account) for the Operations & Management dashboards — including history (e.g. from 1 Mar 2026)	P2 (stretch / pilot)
FR-22	Idempotent "Close My Day" — per-system status, retry of failed systems, no duplicate writes on double-click or refresh (via the <code>external_writes</code> ledger)	P0
FR-23	AutoClock sign-in via Google Workspace OIDC — no passwords stored	P1

P0 = must, P1 = should (hackathon if time / pilot), P2 = stretch, P3 = future. **Phase mapping (per Codex review): P0 = M0** (built at the hackathon) · **P1 = M1** (pilot) · **P2 = stretch** (only if the core is solid by Hour 12) · **P3 = future**. See ERD §14.1.

Jira access model (3 credentials — see ERD §4 / §7): (a) **each employee's own** Jira credential writes *their* worklogs; (b) **one broad "Browse Projects" account** (the Operations member's) reads *all* worklogs for the dashboards; (c) the Atlassian **org admin** key is optional and only imports the user roster — it cannot touch Jira worklogs. AutoClock needs **no Clockwork access** — Clockwork merely displays native Jira worklogs.

9. Error States & Edge Cases

Case	Expected behaviour
Jira worklog POST returns 403 (no permission on ticket)	Mark that entry failed with a clear message; do not block the others; surface "ask your lead for issue access"
Jira / Google API rate-limited (429)	Retry with exponential backoff + jitter; if still failing, queue for retry and tell the user
User's OAuth token expired	Prompt the user to reconnect; never silently drop data
Overlapping time slots (e.g., 9:00–10:00 and 9:30–10:30)	Flag the overlap in the preview; let the user confirm or fix
Total logged > 24h in a day	Block "Confirm" with a validation message
"1.5h" / "90m" / "1h30" durations	Deterministic parser normalizes all to minutes
Jira is down during EOD sync	Queue the worklogs; retry; the Sheet + Gmail still proceed
Google Sheet has formulas/merged cells	Append only to the defined data range; validated against a copy first
Employee on approved leave	Excluded from the Operations chase for those days

Network offline (extension)

M0: popup shows a clear "offline — not saved" state. **M1:** local `chrome.storage` queue + reconciliation on reconnect (ERD §14.6)

10. Privacy, Security & Compliance

- **Per-user authentication** — each employee connects their own Jira and Google accounts; AutoClock never uses one shared bot account (see ERD §7). Worklogs are correctly attributed; data access stays within each user's own permissions.
- **No surveillance** — no screenshots, no keystroke logging, no background activity capture. AutoClock only stores what the employee explicitly types.
- **Data minimization** — store only worklog entries, durations, descriptions, and the per-user tokens needed to write to Jira/Google.
- **Secrets** — all tokens encrypted at rest; environment variables only; never committed.
- **RBAC** — employees see only their own data; leads see their team; management sees aggregates; operations sees hours (not ticket detail); admin configures.
- **Self-hosted** — runs on an Iksula internal server; no employee data leaves the Iksula network.
- **Retention** — worklog data kept for the period Iksula's policy requires; a user's data is removable on request.

11. Accessibility

Target **WCAG 2.1 AA basics**: keyboard-navigable forms and buttons; visible focus states; colour contrast $\geq 4.5:1$ for text; status never communicated by colour alone (icon + text); readable default font sizes. Full audit is a post-pilot item.

12. Constraints & Assumptions

- **C-1 (hard)**: Zero cost — every component free or already owned.
- **C-2 (hard)**: Built for the real ~60 daily users.
- **C-3**: Hackathon build window is 18 hours.
- **C-4**: Iksula already has Jira (with Clockwork Lite/Free) and Google Workspace — AutoClock adds no new subscription.
- **A-1 (assumption)**: An existing Iksula internal server is available to self-host the backend. (*Confirm with Tejas / IT Infra.*)
- **A-2 (resolved)**: Jira access confirmed with the admin — hackathon uses a 3-day personal API token; the pilot uses OAuth 2.0 (3LO). The org admin API key is **not** usable for Jira worklogs (admin API only). The Operations member's account provides the broad "Browse Projects" read access for dashboards. Needs **only** the *Browse Projects* + *Work On Issues* permissions.
- **A-3**: $\geq 80\%$ of the 60 use Chrome as their primary browser (extension assumption — to verify).

13. Release / Launch Checklist

Hackathon (Sat 23 May):

- [] Employee log flow works end-to-end on one real user (Yogesh's token).
- [] "Close My Day" writes a real Jira worklog, a real Sheet row, a real Gmail draft.
- [] One role dashboard renders with seeded data.
- [] Backup demo video recorded.
- [] 60-second pitch + dry run done.

- [] Self-hosted on the Iksula server; HTTPS; secrets in env.
- [] Per-user Jira + Google OAuth onboarding built and tested.
- [] Timezone (IST↔UTC) verified against Clockwork.
- [] Leave calendar wired into the Operations target.
- [] Extension force-installed via Google Workspace Admin.
- [] 5–10 users onboarded first, then the rest of the 60.

14. Success Metrics & Pilot

Measured over a 2-week pilot with the ~60 users — see GM-01...GM-05 (§3). The headline: a full day logged in **under 2 minutes**, **zero running cost**, and **≥ 80%** of the 60 still using it after two weeks.

15. Milestones (overview — detail in the Work Plan)

Phase	Outcome
M0 — HackFest (18h)	Working prototype: employee flow + one dashboard, on one user's token
M1 — Pilot rollout (2–4 wks)	Self-hosted, per-user OAuth, hardened, the ~60 onboarded
M2 — Full roles (1–2 mo)	PM / Management / Operations / Admin consoles complete; Fri/Mon cron live
M3 — Optional later	Wider rollout only if leadership asks

16. Open Questions & Risks

ID	Item	Owner
OQ-1	Jira access — <i>partially resolved</i> : model agreed (3-day token → OAuth). Still to confirm before Friday : classic vs scoped token, exact site URL + project keys, issue-level security, reader-account token	Yogesh + Tejas
OQ-2	Which Iksula server hosts the backend	Tejas
OQ-3	Exact structure of the real Google Sheet timesheet (tab name, columns, formulas)	Ravi
OQ-4	The current EOD email template (recipients, subject, HTML)	Yogesh
OQ-5	Leave/holiday data source for the Operations target	Tejas / HR
OQ-6	Google Cloud project + Internal OAuth consent screen + demo refresh token created before Friday	Ravi

